

Seeded LLM Tournament

A 5-Step Methodology for High-Precision Information Retrieval & Re-ranking

Ron Dagani Gilad Tsfaty Tal Hibner

Method 1 : Hybrid Multi-Stage Ranking with Reciprocal Rank Fusion (RRF) Cormack et al. 09

METHODOLOGY COMPONENTS

LEXICAL RETRIEVAL

BM25

Robust Baseline

Role: Primary first-stage retriever

Parameters: k1=0.9, b=0.4

Note: Tuned specifically for neural reranking synergy

QUERY EXPANSION

RM3 PRF

Pseudo-Relevance Feedback

Role: Improves recall via expansion

Config: fb_docs=15, fb_terms=120

Weight: original_query_weight=0.03

NEURAL RERANKING

MonoT5-large

Seq2Seq Transformer

Role: Deep semantic understanding

Model: castorini/monot5-large-msmarco-10k

Scale: 770M parameters

FUSION STRATEGY

Reciprocal Rank Fusion

Combines signals from BM25, RM3, and MonoT5 using rank-based weighting.

$$RRFscore(d) = \sum(r \in R) (1 / (k + rank(d)))$$

Configuration:

k = 10 (Optimized for this setup, typically 60)

3-way Fusion: Equal weighting

Depth: Top 1000 documents

Findings & Performance Analysis



Parameter Sensitivity

BM25 parameters required significant retuning when upgrading to MonoT5-large. Optimal values shifted from standard (0.7, 0.6) to ($k_1=0.9$, $b=0.4$), indicating neural rerankers prefer different lexical candidates.



Aggressive Query Expansion

A very low original query weight (0.03) in RM3 provided the best results. The strong neural reranker effectively filters noise introduced by aggressive expansion, maximizing recall.



Reranker Impact

Scaling from MonoT5-base to MonoT5-large yielded a substantial ~8% MAP improvement. The 770M parameter model demonstrates superior semantic understanding for complex topics.



Fusion Tuning (RRF)

For our 3-way fusion setup, a lower constant of $k=10$ outperformed the paper's recommended $k=60$. This emphasizes the top-ranked documents more aggressively in the final score.



"Less is More"

Adding SPLADE or Contriever actually degraded performance (-2.2% and -1.8%). Their signals overlapped with the existing pipeline, introducing noise rather than complementary information.

Challenges Encountered



Pyserini Compatibility

Version discrepancies in Pyserini APIs and prebuilt index availability required strict package management.



Compute Constraints

MonoT5-large inference requires ~ 130 s per query on Colab T4 GPU, limiting hyperparameter search space.



Diminishing Returns

Adding advanced signals like SPLADE and Contriever unexpectedly degraded performance (-1.8% to -2.2%).

Method 2 : The Seeded LLM Tournament

CORE PHILOSOPHY

- Maximize early precision with LLM reasoning capabilities
- Maintain long-tail ranking stability using classical IR
- Balance computational cost with token-efficient strategies

PROCESSING PIPELINE



Data Processing & Query Expansion

Preparing the lexical foundation for retrieval models

PROCESS A

Aggressive Cleaning

Apply:

Normalization: Standardize case (lower) and Unicode characters.

Stripping: Remove HTML markup, boilerplate text, and system artifacts.

Whitespace: Collapse multiple spaces and line breaks into single tokens.

Denoising: Filter out non-alphanumeric noise patterns in narrative fields.

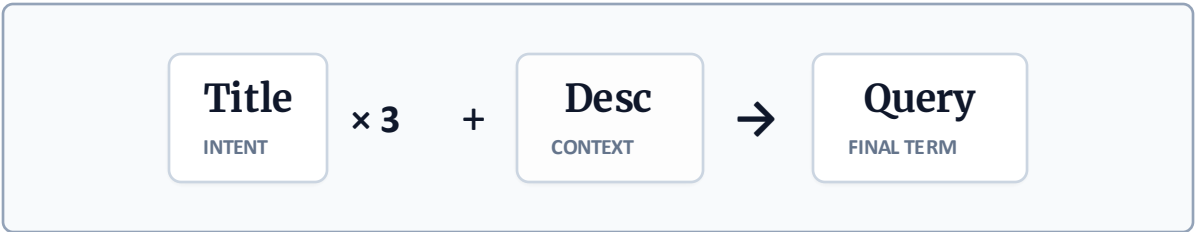
IMPACT

Reduces index size by ~15% and prevents "garbage-in-garbage-out" matching in the vector space.

PROCESS B

Weighted Query Construction

Rationale: The Title represents core intent, while the Description provides context but adds noise.



WITHOUT WEIGHTING

Query: "server crash error log analysis for linux system"

WITH WEIGHTING (3X)

Query: "server crash server crash server crash error log..."

Mechanism: Tripling the title ensures BM25 term frequency (TF) and Query Likelihood models prioritize the primary search terms over verbose descriptions.

Base Retrieval Fusion

Combining probabilistic and language models for a robust recall foundation

MODELS

Hybrid Retrieval Setup

BM25 (PROBABILISTIC)

Primary retriever known for robustness.

k1 = 1.2 // Saturation
b = 0.75 // Length norm

QUERY LIKELIHOOD (LM)

Provides smoothing based on doc probability.

$\mu = 1000$ // Dirichlet smoothing

WHY HYBRID?

BM25 handles term matching effectively, while QL captures the likelihood likelihood of generating the query from the document, offering complementary signals.

FUSION

Expansion & Scoring Strategy

RM3 EXPANSION (RELEVANCE MODEL V3)

Addresses vocabulary mismatch by expanding the query with semantically related terms.

30
TERMS ADDED

Top 10
DOCS ANALYZED

"Adds synonyms without manual synonyms list"

SCORE FUSION LOGIC

$$\underset{W_{BM25}}{0.8} + \underset{W_{QL}}{0.2} \rightarrow \underset{INITIAL RANKING}{\text{Base Score}}$$

BM25 acts as the robust anchor (80%), while QL provides a 20% smoothing effect. This weighted combination ensures high recall and stability before the expensive LLM reranking stage.

Semantic Passage Extraction- on top 300 docs

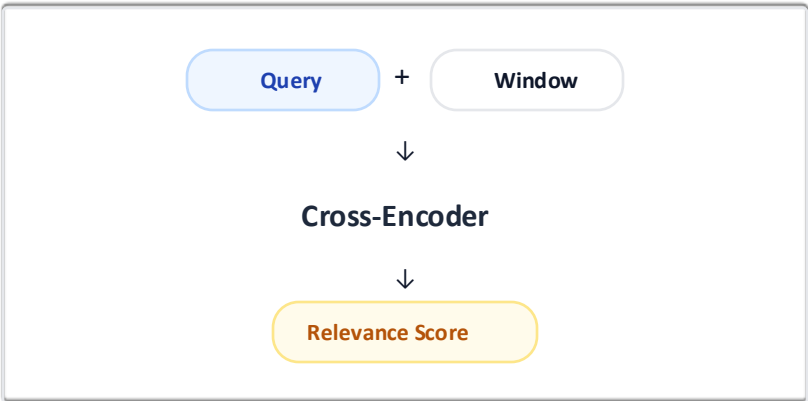
Concentrating evidence for high-precision judging using Cross-Encoders

MODEL ARCHITECTURE

Cross-Encoder Pairing

model: cross-encoder/ms-marco-MiniLM-L-6-v2

Mechanism: Performs Input Pairing. The query and a text window are fed into a single transformer simultaneously to allow full token interaction.



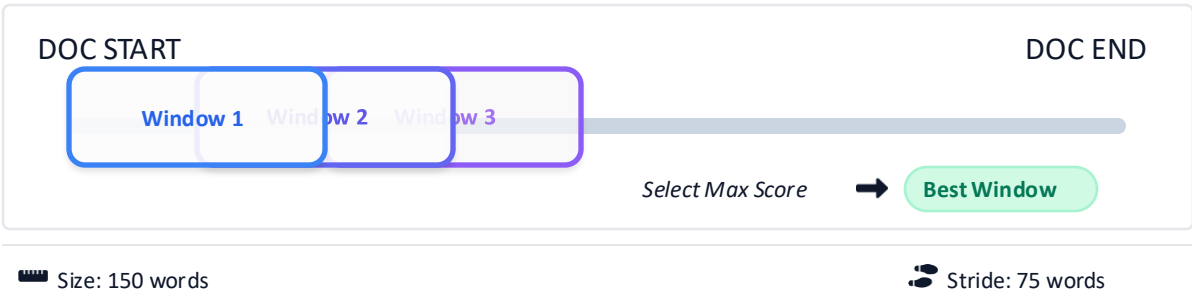
WHY USE THIS?

Analyses deep semantic relationships and patterns rather than just keyword matching.

STRATEGY

Sliding Window Selection

LLMs have token limits. We break documents into overlapping chunks to identify the most concentrated evidence.



INPUT FILTERING

Ensures the LLM judges only the most relevant section, preventing distraction by noise in long documents.

EFFICIENCY

Reduces token usage significantly compared to feeding full documents, enabling faster processing.

The Seeded Tournament

List-wise LLM judging - "Seeded"(e.g.,1st vs.300th,299th,and 298th).

MECHANISM



List-Wise Judging

Model: gpt-4o-mini
Mode: List-Wise (Batch of 4)

Instead of pair-wise comparisons, the LLM evaluates 4 documents simultaneously. This optimizes token efficiency while leveraging the model's ability to rank items relatively.

Seeded Bracket: Top-ranked documents from base retrieval ("Seeds") are matched against lower-ranked ones. Then in later rounds a "Random" seeding, to allow the LLM to judge diverse pairings.

Tournament Normalization: Min-Max Normalization (divided by Max-Score).

SCORING FORMULA

$10 \times 2^{(\text{round}-1)}$

Exponential points reward documents that survive multiple rounds, pushing winners to the absolute top.

BRACKET VIEW



Seeded Competition Flow

ROUND 1
-300 docs
seeded

MATCH A
4 Docs

MATCH B
4 Docs

MATCH C
4 Docs

MATCH D
4 Docs

ROUND 2 -75
docs random

SEMI-FINAL 1
4 Winners

SEMI-FINAL 2
4 Winners

FINAL

CHAMPION
Top Rank



Winners of each batch advance to face other winners, concentrating relevance at each step.

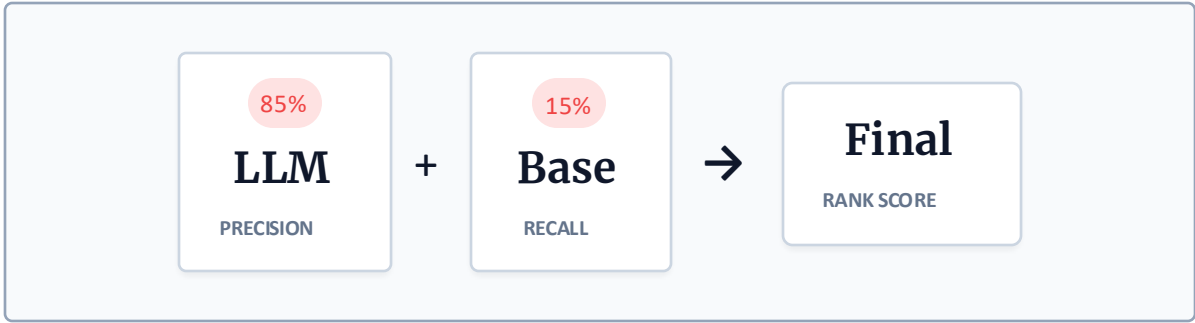
Grand Fusion

Combining innovation with stability for the final ranking

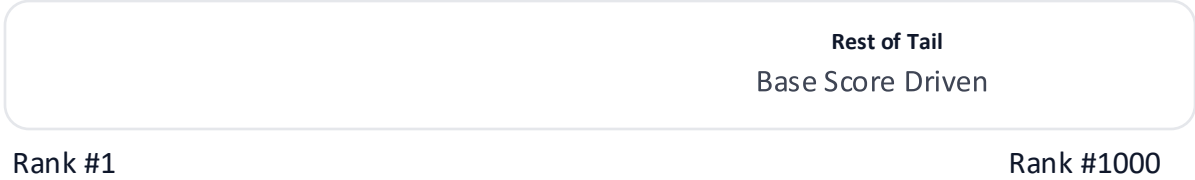
LOGIC

Weighted Combination Strategy

The final ranking score is a linear combination of the LLM Tournament outcome and the robust Base Retrieval foundation.



RANKING COMPOSITION



RATIONALE

Why This Balance?

Innovation Driver (85%): The LLM is significantly better at deep semantic understanding and identifying high-precision results at the very top.

Safety Anchor (15%): Traditional IR (BM25/QL) is more stable for general ranking. It acts as a tie-breaker when the LLM is unsure.

Tail Management: For documents not reached by the tournament (beyond top 300), the base score ensures a sensible ordering remains.



FINAL OUTCOME

Maximize MAP (Mean Average Precision) by letting "Innovation" drive the top ranks while "Tradition" secures the foundation.



**THANK YOU
FOR LISTENING**